Structural Stability

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AISC-LRFD Design Curve

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Euler Buckling Load:

$$P_{\mathrm{cr}} = \frac{\pi^2 EI}{(KL)^2}$$

Buckling Strength (탄성좌굴강도):

$$F_{\mathrm{cr}} = \frac{\pi^2 EI}{A(KL)^2} = \frac{\pi^2 E}{(KL/r)^2}$$

where r: 단면2차반경

$$r = \sqrt{I/A}$$

항복강도(F_y)와 탄성좌굴강도(F_{cr})의 교차점: $F_y = F_{\mathrm{cr}}$에 대한 KL/r 값

$$F_{\mathrm{cr}} = F_y \Rightarrow \frac{\pi^2 E}{(KL/r)^2} = F_y \Rightarrow (KL/r)_i = \pi\sqrt{\frac{E}{F_y}}$$

AISC-LRFD Design Curve and KDS Formulation

Dimensionless Slenderness Ratio (무차원세장비):

$$\lambda_c = \frac{KL}{r}\sqrt{\frac{F_y}{\pi^2 E}}$$

Design Curve:

$$\frac{F}{F_y} = \begin{cases} \exp(-0.419\lambda_c^2) & \lambda_c \leq 1.5 \\ 0.877\lambda_c^{-2} & \lambda_c > 1.5 \end{cases}$$

KDS Formulation:

$$\lambda_c = \frac{KL}{r}\sqrt{\frac{F_y}{\pi^2 E}} = \sqrt{\frac{F_y}{F_e}} \leq 1.5 \Rightarrow$$